Predicting 30-Day Hospital Readmission for Diabetic Patients

Phase 2: Classification Modeling & Comparative Analysis

Roadmap

1. Phase 1 Recap
2. Experimental Setup
3. Models (5)
4. Comparison
5. Explainability
6. Fairness
7. Discussion & Limitations
8. Clinical Use, Conclusion

Phase 2 Focus

Moving from features to insights: evaluating **5 models**, assessing **fairness**, and discussing clinical applicability.

Phase 1 Recap (High-Level)

101K
Raw Encounters

↓

71K
Unique Patients

↓

~70K
Clean Data Rows

↓

98 Features
Engineered & Selected

Data Characteristics

9% Positive Class

Severe class imbalance requiring specific weighting strategies across all classification models.

Key Predictors Identified

Prior hospitalization metrics and discharge disposition showed strong early predictive signals.

98 Features Retained

Encompassing healthcare utilization, patient demographics, clinical diagnoses, and medications.

Experimental Setup & Methodology

Dataset Partitioning & Handling

Full Dataset (~70,000) - ~9% Positive

Training Set (80%)
Stratified Split (random_state=42)

Test (20%)
Hold-out

Imbalance Handling Strategy

Consistent algorithmic approach across all models:

- Scikit-learn: `class_weight='balanced'`
- XGBoost: `scale_pos_weight ≈ 10.15`
- Keras: Custom class_weight dictionary

Evaluation Metrics

Primary: AUC-ROC
Threshold-free, robust to severe class imbalance

Secondary Metrics (threshold = 0.5)
F1 Score, Precision, Recall

The Accuracy Paradox
Predicting all zeros yields ~91% accuracy, rendering it useless for clinical decision making.

Hyperparameter Tuning

Method: 5-fold Stratified Cross-Validation on training set

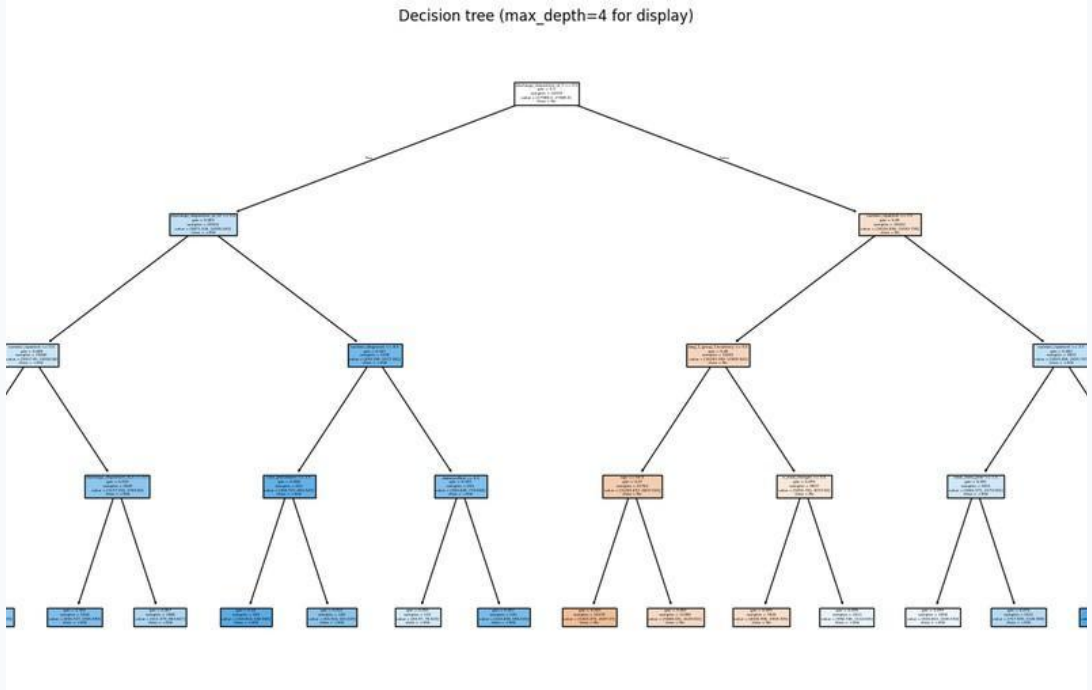
- `GridSearchCV`: Decision Tree, Logistic Regression, SVM
- `RandomizedSearchCV`: XGBoost

Fair Comparison

All 5 models evaluate on the same 20% stratified test set, using consistent imbalance handling strategies and evaluated by the same primary metric (AUC-ROC).

Model 1 — Decision Tree

Decision Tree Visualization (Depth 4 Display)



Best Hyperparameters (GridSearchCV)

max_depth: 7
min_samples_leaf: 60
min_samples_split: 2

Test Metrics (Threshold = 0.5)

AUC-ROC:**0.6152**
F1 Score:**0.2008**
Acc:**0.5864**
Prec:**0.1214**
Recall:**0.5793**

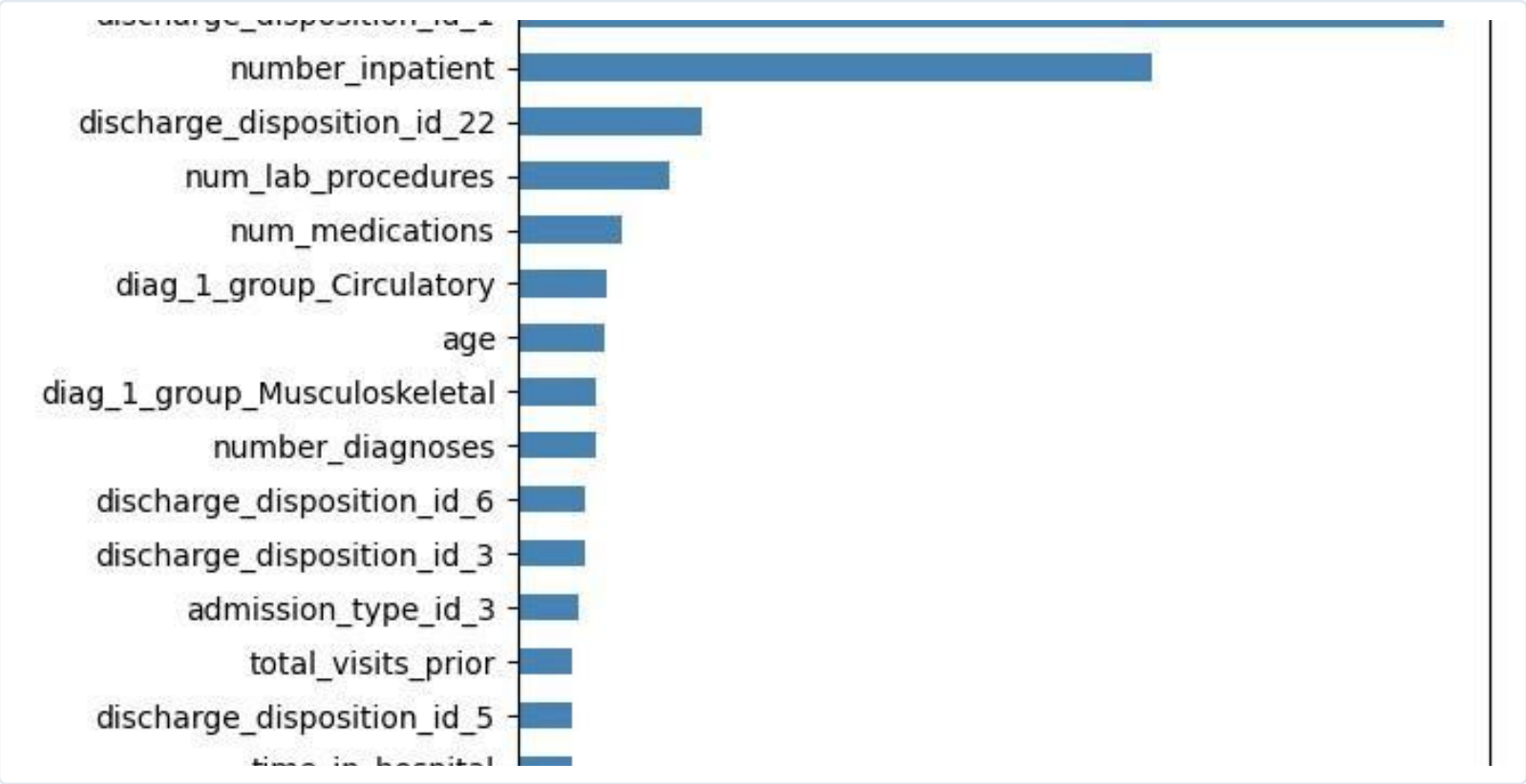
Overfitting Diagnostic

Default tree overfits badly (0.5209 AUC) — pruning was critical.
Train-test AUC gap reduced to **0.049**.

Key Takeaway

Highly interpretable architecture, but proves to be the **weakest ranker** among evaluated models.

Decision Tree — Feature Importance



Key Drivers

discharge_disposition_id_1 overwhelmingly dominates the model's split decisions and risk assessment.

number_inpatient (prior utilization) serves as the second strongest predictor for readmission.

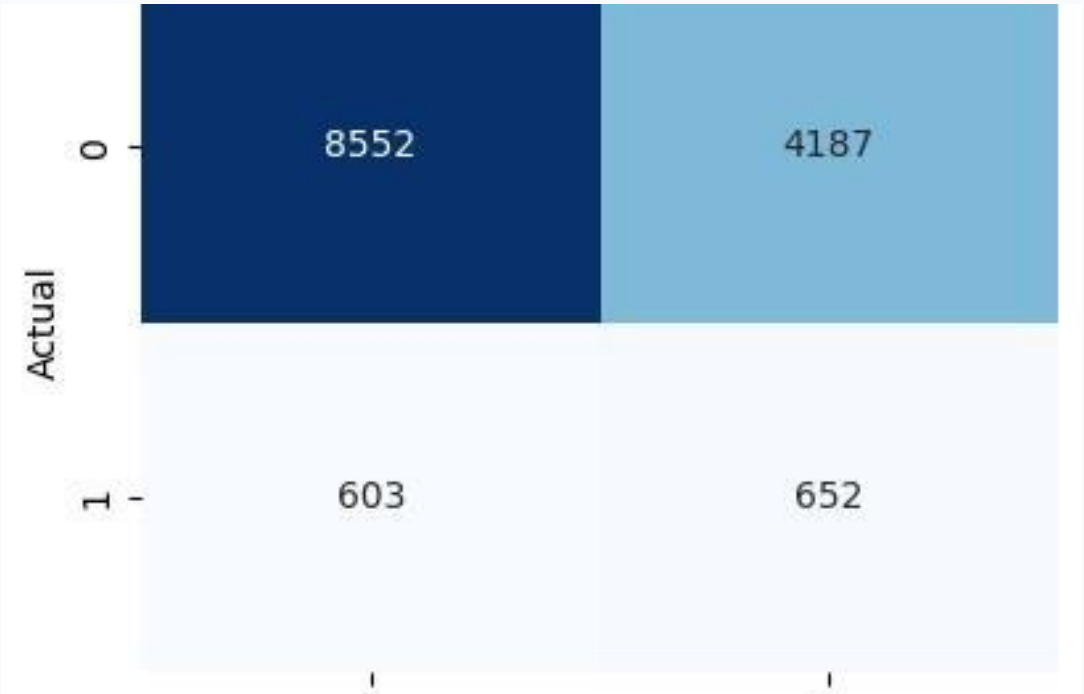
These top features directly align with standard clinical intuition regarding hospital readmission risks.

Key Takeaway

The Decision Tree relies heavily on discharge context and past utilization, matching real-world clinical risk assessments.

Model 2 — Logistic Regression

Confusion Matrix



Best Hyperparameters (GridSearchCV)

C: 0.001

Note: Strong regularization preferred

Test Metrics (Threshold = 0.5)

AUC-ROC:**0.6444**

F1 Score:**0.2140**

Acc:**0.6577**

Prec:**0.1347**

Recall:**0.5195**

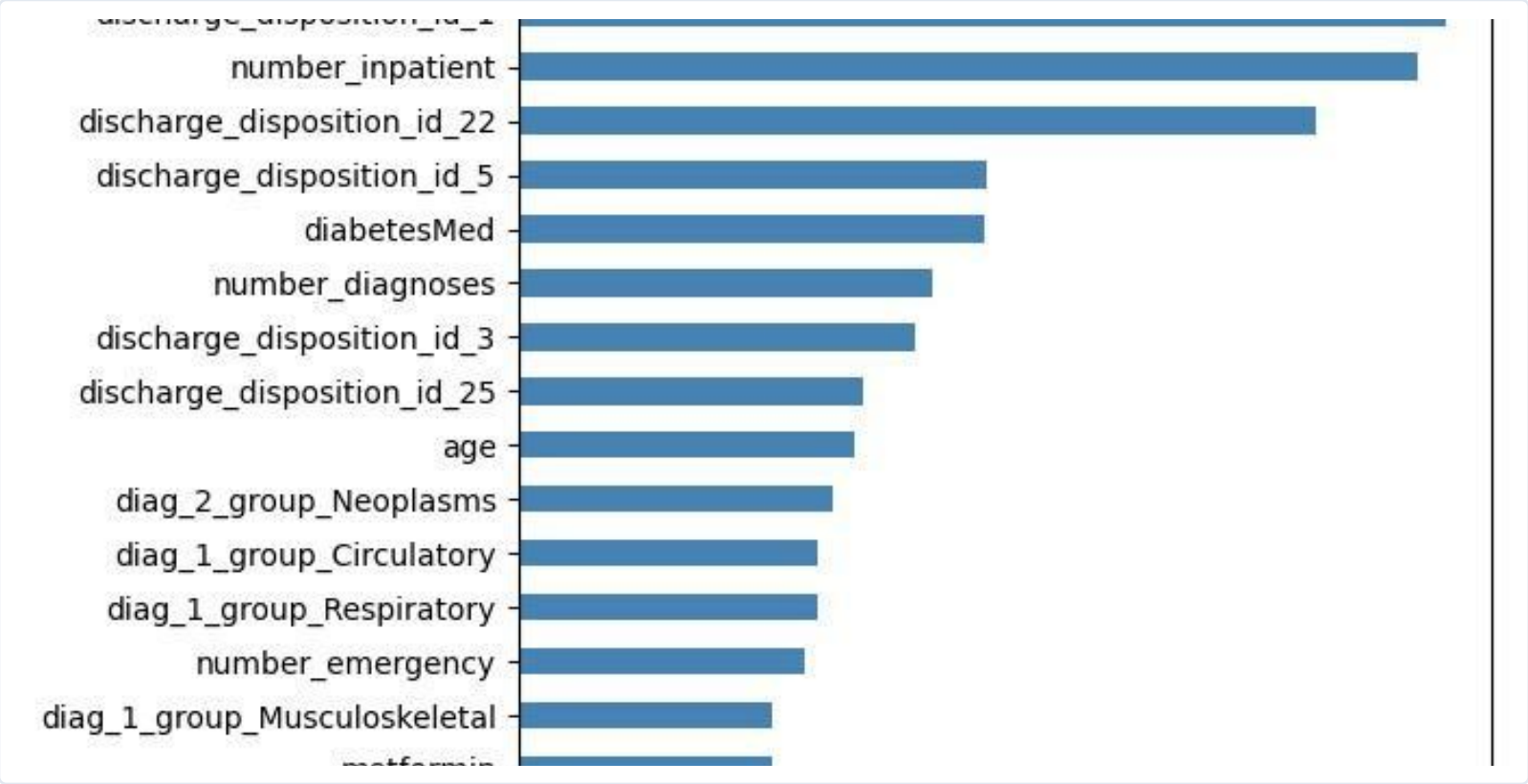
Performance Note

Train-test gap is **0.013** (very stable, low variance). Improvement over DT: **+0.03 AUC** — linear baseline is competitive.

Key Takeaway

A **strong, auditable baseline** where coefficients are directly interpretable by clinicians.

Logistic Regression — Coefficients



Coefficient Interpretation

Coefficients represent the change in log-odds of readmission per standardized unit change in the feature.

This direct relationship is highly auditable by clinicians, allowing them to verify clinical validity.

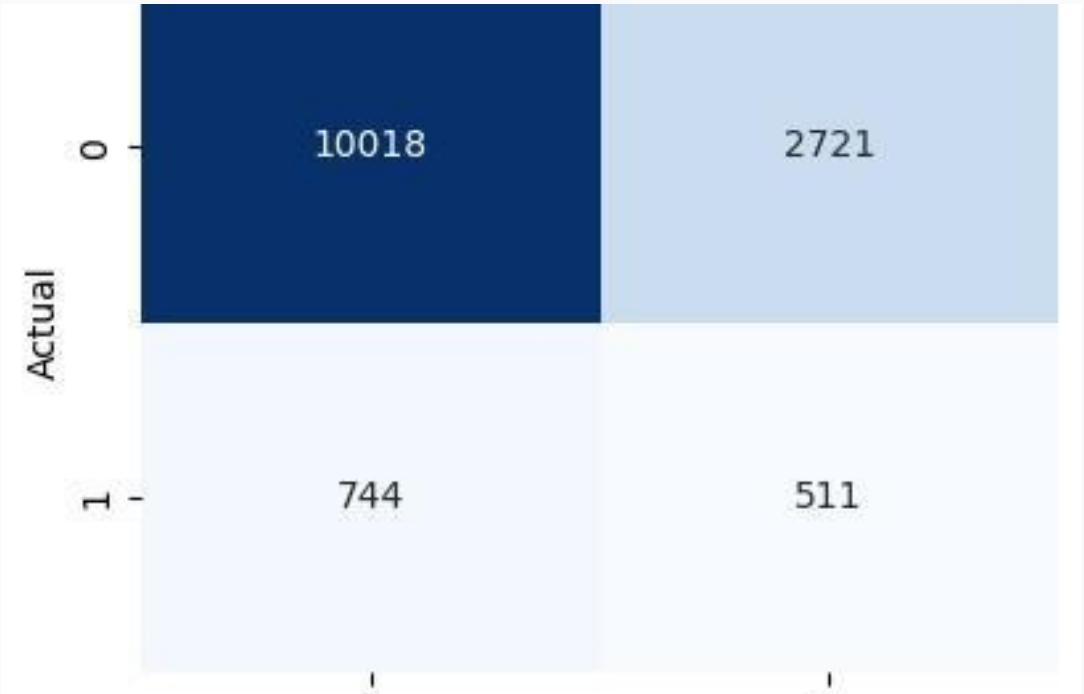
Top features align strongly with known clinical risk factors like prior utilization.

Key Takeaway

Logistic Regression provides transparent, magnitude-based feature importance that bridges the gap between statistical modeling and clinical intuition.

Model 3 — SVM (Linear)

Confusion Matrix (Threshold = 0.108)



Model Setup & Hyperparameters

Why LinearSVC: RBF kernel is $O(n^2)$ memory, impractical for 56K rows × 98 features.

Best C: 0.001

Test Metrics (Threshold = 0.108)

AUC-ROC:**0.6442**

F1 Score:**0.2278**

Acc:**0.7524**

Prec:**0.1581**

Recall:**0.4072**

Calibration Issue

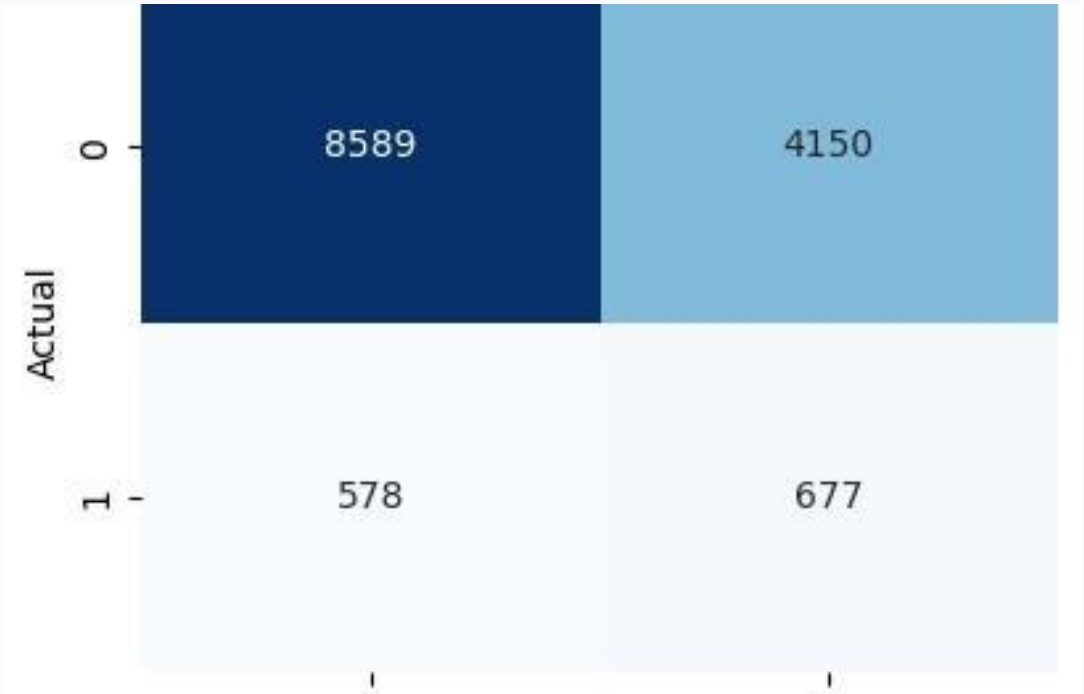
Calibrated model at 0.5 threshold predicted ~1 positive out of 13,994. A threshold shift to **0.108** recovered meaningful predictions.

Key Takeaway

A **good ranker**, but deployment needs explicit thresholding to correct for calibration on highly imbalanced data.

Model 4 — XGBoost

Confusion Matrix



Best Hyperparameters (RandomizedSearchCV)

n_estimators: 500
max_depth: 5
lr: 0.01
subsample: 0.7
colsample: 0.7
scale_pos_weight: ≈10.15

Test Metrics (Threshold = 0.5)

AUC-ROC:	**0.6505 (BEST)**
F1 Score:	**0.2226**
Acc:	**0.6621**
Prec:	**0.1403**
Recall:	**0.5394**

Overfitting Diagnostic

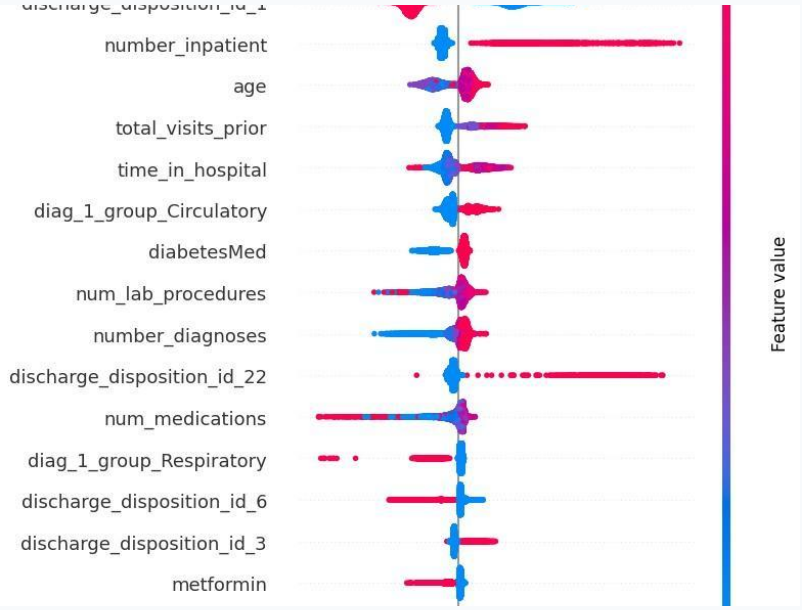
Train-Test AUC gap: **0.066** (Moderate overfit despite regularization).

Key Takeaway

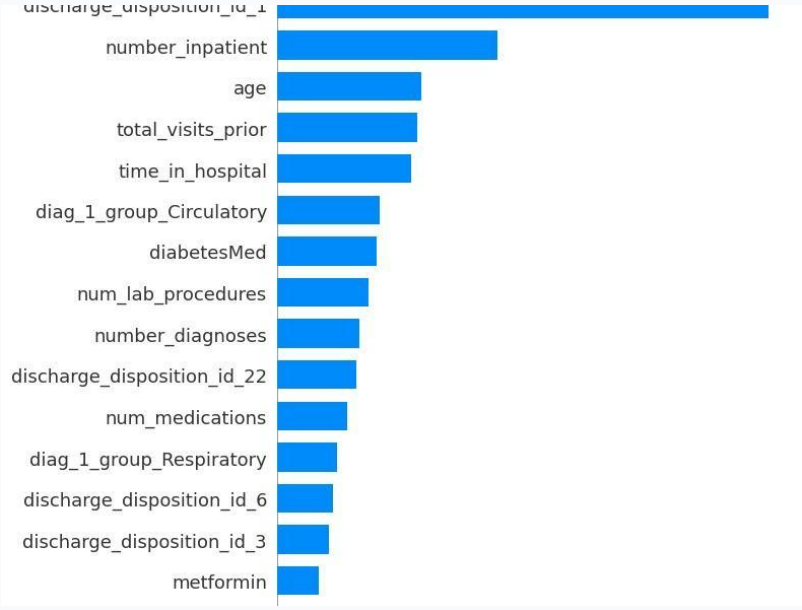
XGBoost serves as the **best overall ranker**, confirming its typical dominance on tabular datasets.

XGBoost — SHAP Explainability

SHAP Beeswarm (Direction + Magnitude)



Mean |SHAP| (Feature Importance)



Top SHAP Values

discharge_disposition_id_1: 0.227

number_inpatient: 0.102

age: 0.067

The magnitude of **discharge_disposition** is over 2× the next feature, showing its massive influence.

Clinical Interpretation

Prior utilization history and the discharge context **overwhelmingly dominate readmission risk**.

Model 5 — Neural Network

Architecture Comparison

Model A (Basic) - *Winner*

```
Input → Dense(64) → Dropout(0.3) → Sigmoid
```

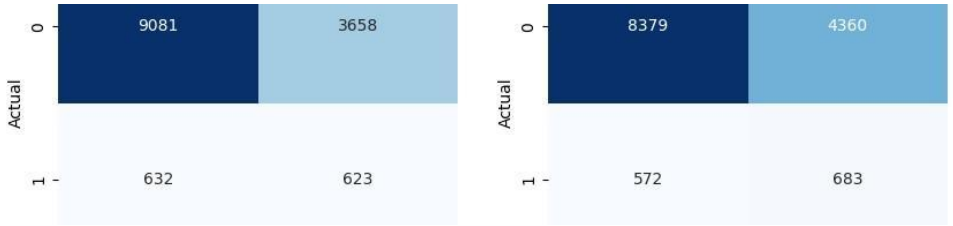
15 epochs (early stop at 20)
Test AUC: **0.6443**

Model B (Deep) - *Overtrained*

```
Input → Dense(128) → BN → Drop →
Dense(64) → BN → Drop → Dense(32) → Drop → Sigmoid
```

50 epochs
Test AUC: **0.6383** (worse than basic)

Confusion Matrices (Basic vs Deep)



Test Metrics (Basic Model, Threshold = 0.5)

AUC-ROC:**0.6443**
F1 Score:**0.2251**
Acc:**0.6934**
Prec:**0.1455**
Recall:**0.4964**

Overfitting Diagnostic

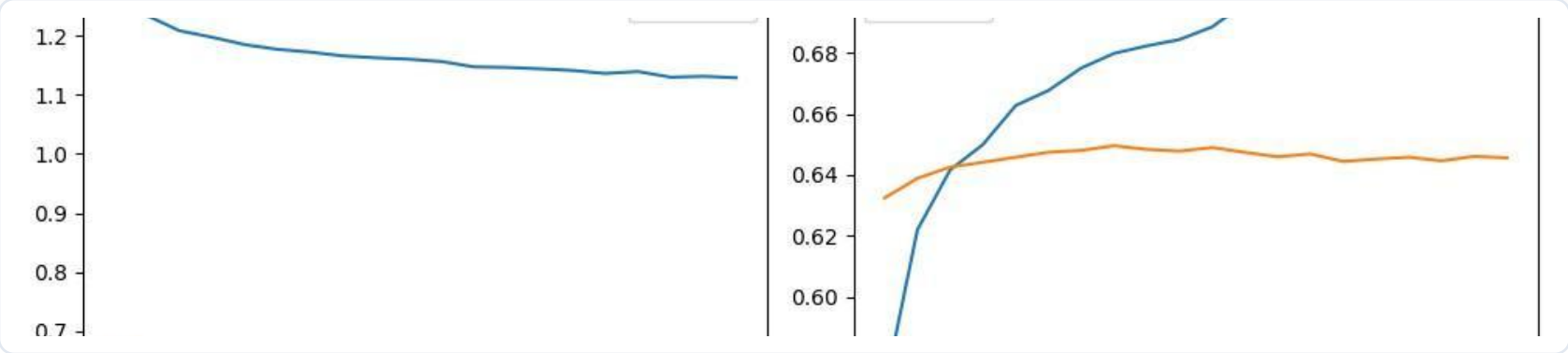
Deep model overtrained severely with a train-test AUC gap of **0.156** vs **0.079** for the basic model.

Key Takeaway

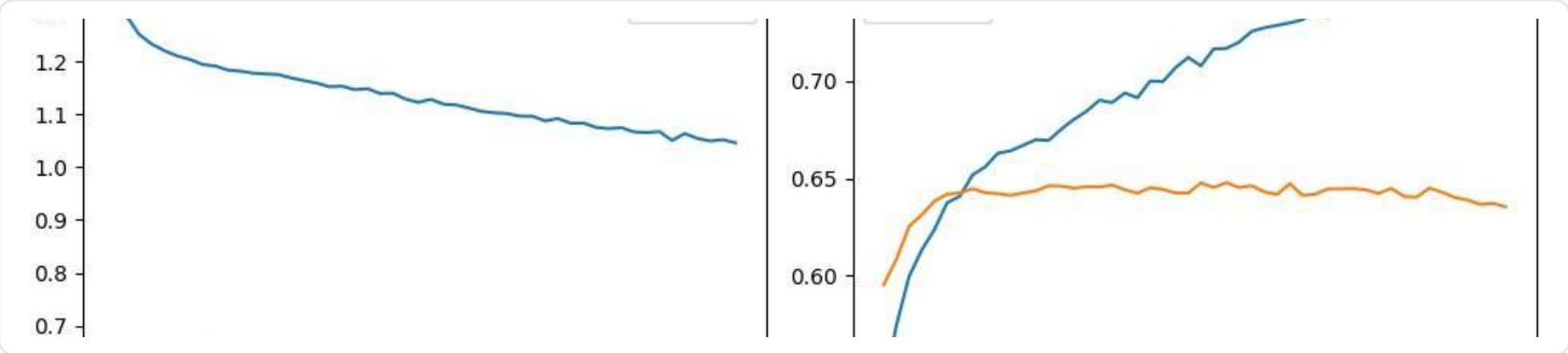
~70K rows and 98 hand-crafted features is **not** the regime where deep networks shine — boosted trees are the industrial default here.

Neural Network — Learning Curves

Basic Architecture



Deep Architecture



Key Observations

- Basic model validation AUC plateaus early (~epoch 7).
- Deep model shows significant divergence between train and validation (overfitting).
- Early stopping prevented further degradation but deeper architecture failed to generalize.

Key Takeaway

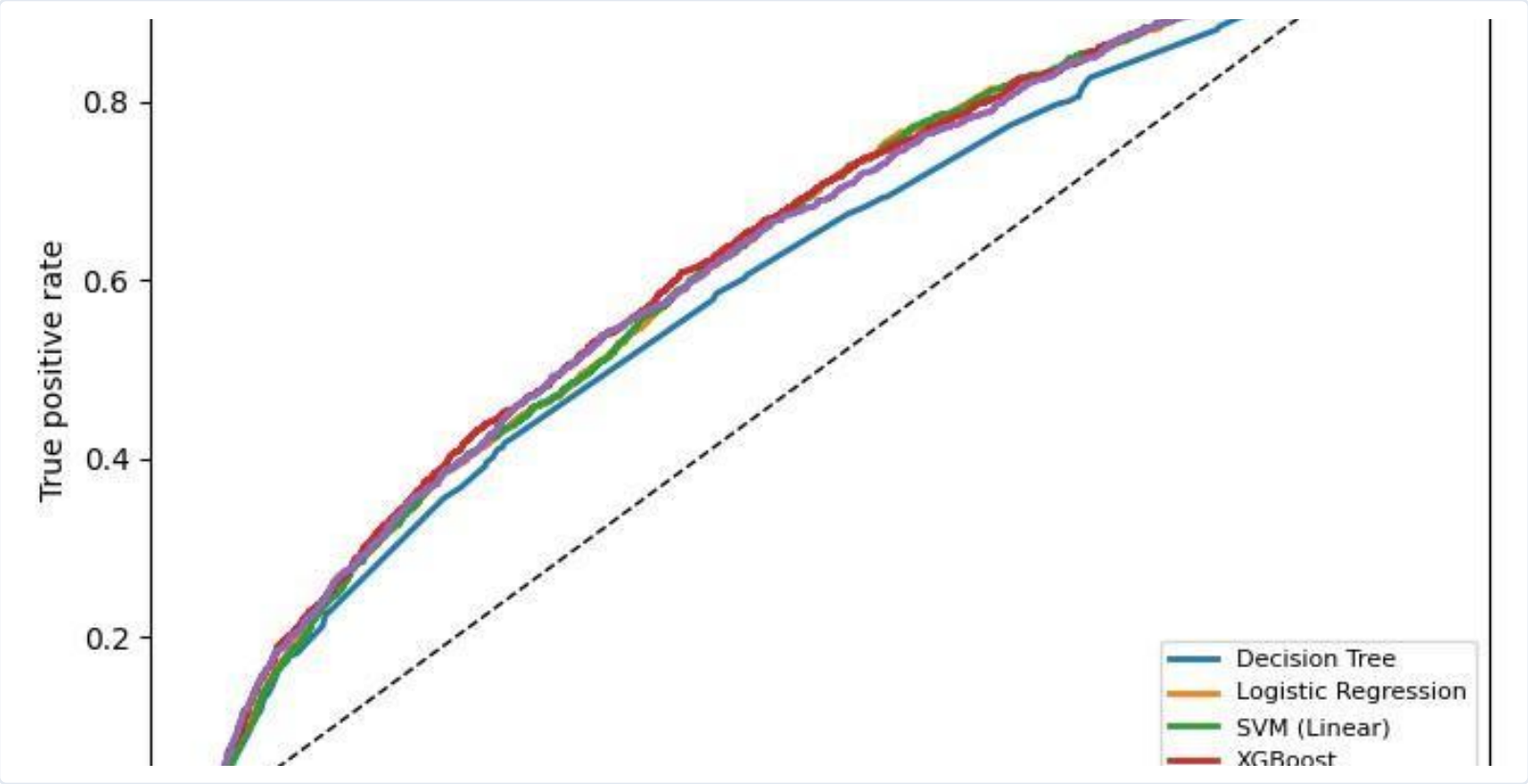
~70K rows and 98 hand-crafted features is not the regime where deep networks shine — boosted trees are the industrial default here.

Model Comparison: Summary

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Decision Tree	0.5864	0.1214	**0.5793**	0.2008	0.6152
Logistic Regression	0.6577	0.1347	0.5195	0.2140	0.6444
SVM (Linear)*	**0.7524**	**0.1581**	0.4072	**0.2278**	0.6442
XGBoost	0.6621	0.1403	0.5394	0.2226	**0.6505**
Neural Net (basic)	0.6934	0.1455	0.4964	0.2251	0.6443

* SVM Linear results shown at an optimized threshold of 0.108. All other models evaluated at default threshold 0.5.
Green indicates best in metric. **Red** indicates worst in metric.

Model Comparison — ROC Overlay



Visual Analysis

Clustered Performance

The ROC curves for all five models cluster tightly within the 0.61 to 0.65 AUC range.

Weak Signal: This tight clustering, far below a "perfect" right-angle curve, visually confirms that the readmission signal in this administrative dataset is inherently weak and noisy.

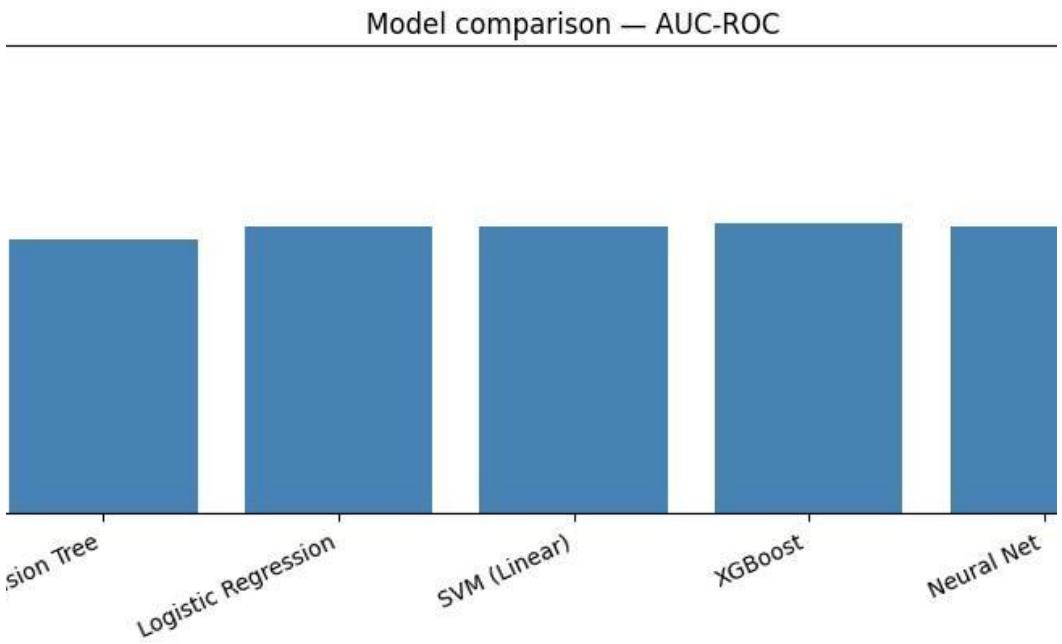
Diminishing Returns: The gap between the simplest model (Decision Tree) and the most complex (XGBoost) is visually small, highlighting the limits of algorithm selection alone.

Key Takeaway

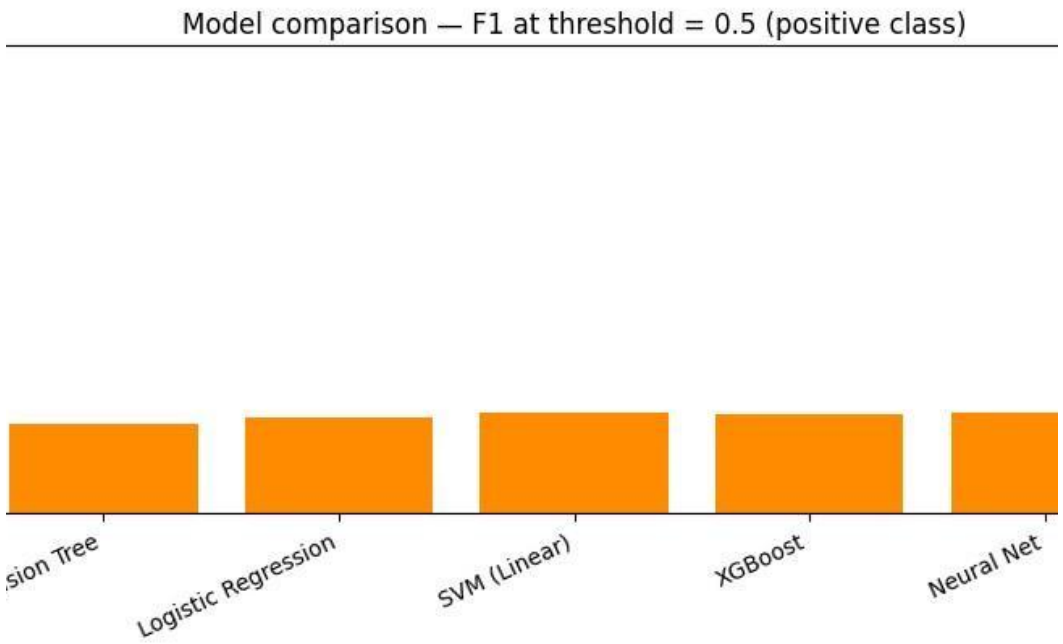
The tight clustering of ROC curves tells a clear story: predicting readmission is inherently difficult with administrative data, and no algorithm offers a magic bullet.

Model Comparison — AUC-ROC vs F1 Score

AUC-ROC (Threshold-Free)



F1-Score (at default threshold 0.5)



Key Insight

While AUC-ROC measures ranking quality robustly against imbalance, F1-score depends entirely on the chosen decision threshold. The default 0.5 threshold yields uniformly low F1 scores across all models.

Overfitting Diagnostic

Model	Train AUC	Test AUC	Gap	Notes
Logistic Regression	0.657	**0.644**	**0.013**	Most stable, highly regularized
SVM (Linear)	0.657	**0.644**	**0.013**	Low variance, matches LR
Decision Tree	0.664	**0.615**	**0.049**	Pruning (depth 7) prevented massive overfit
XGBoost	0.716	**0.650**	**0.066**	Moderate overfitting despite regularization
Neural Net (Basic)	0.723	**0.644**	**0.079**	Early stopping kicked in early
Neural Net (Deep)	0.794	**0.638**	**0.156**	Memorized training set, diverged

Bias-Variance Tradeoff

Simpler, linear models generalize better on this tabular dataset, while high-capacity deep networks overfit severely.

Feature Importance Across Models

Rank	Decision Tree	Logistic Regression	SVM (Linear)	XGBoost
1	discharge_disposition_id_1	discharge_disposition_id_1	discharge_disposition_id_1	discharge_disposition_id_1
2	number_inpatient	number_inpatient	number_inpatient	discharge_disposition_id_22
3	discharge_disposition_id_22	discharge_disposition_id_22	discharge_disposition_id_22	number_inpatient
4	num_lab_procedures	discharge_disposition_id_5	discharge_disposition_id_5	discharge_disposition_id_3
5	num_medications	diabetesMed	diabetesMed	total_visits_prior

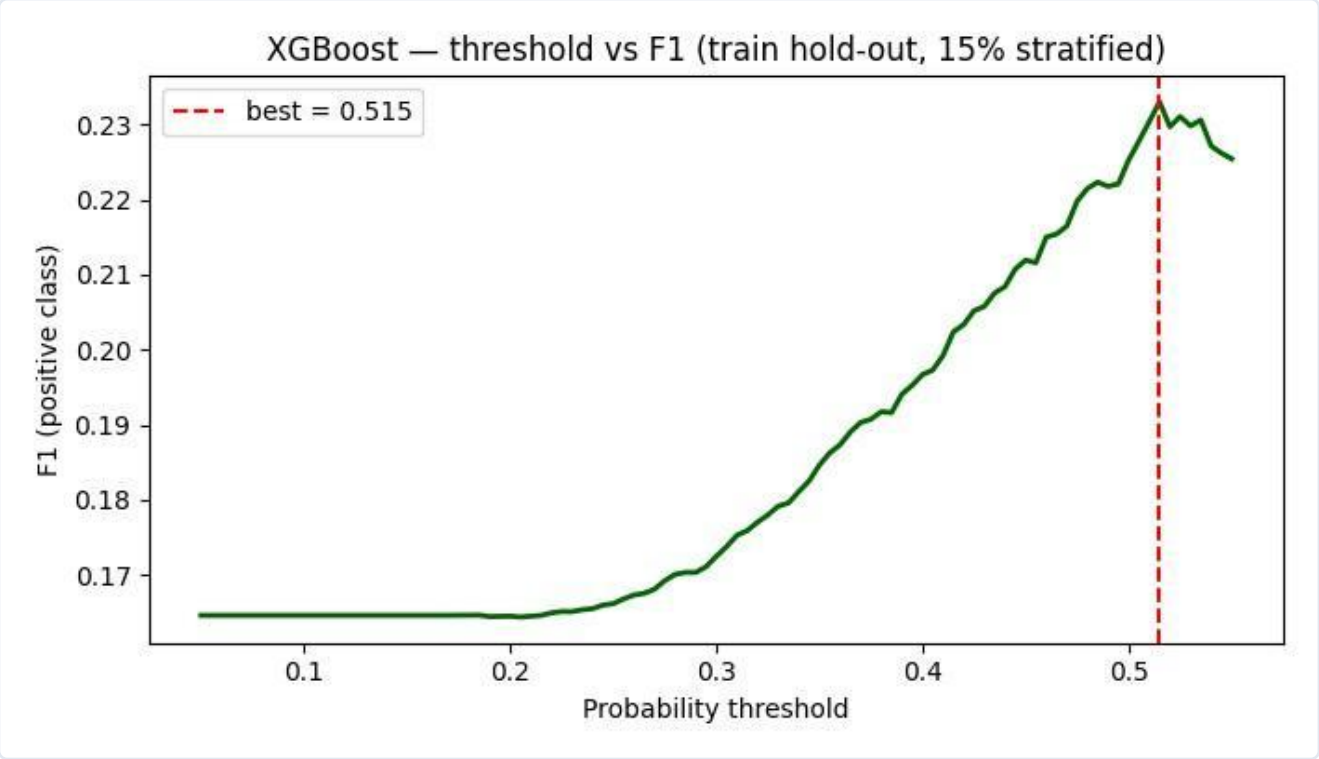
6/10 Overlap between DT and XGBoost

Consistent Top Features: Prior utilization & discharge context

Key Takeaway

Discharge disposition (discharge_disposition_id_1) is the #1 predictor across ALL models. The models tell a highly consistent clinical story.

Threshold Optimization



XGBoost Threshold Sweep

Optimal Threshold

RESULT: 0.515

Default F1 (0.50): 0.2226

Optimized F1 (0.515): 0.2257

Observation: Small delta indicates XGBoost is already well-calibrated.

Clinical Framing

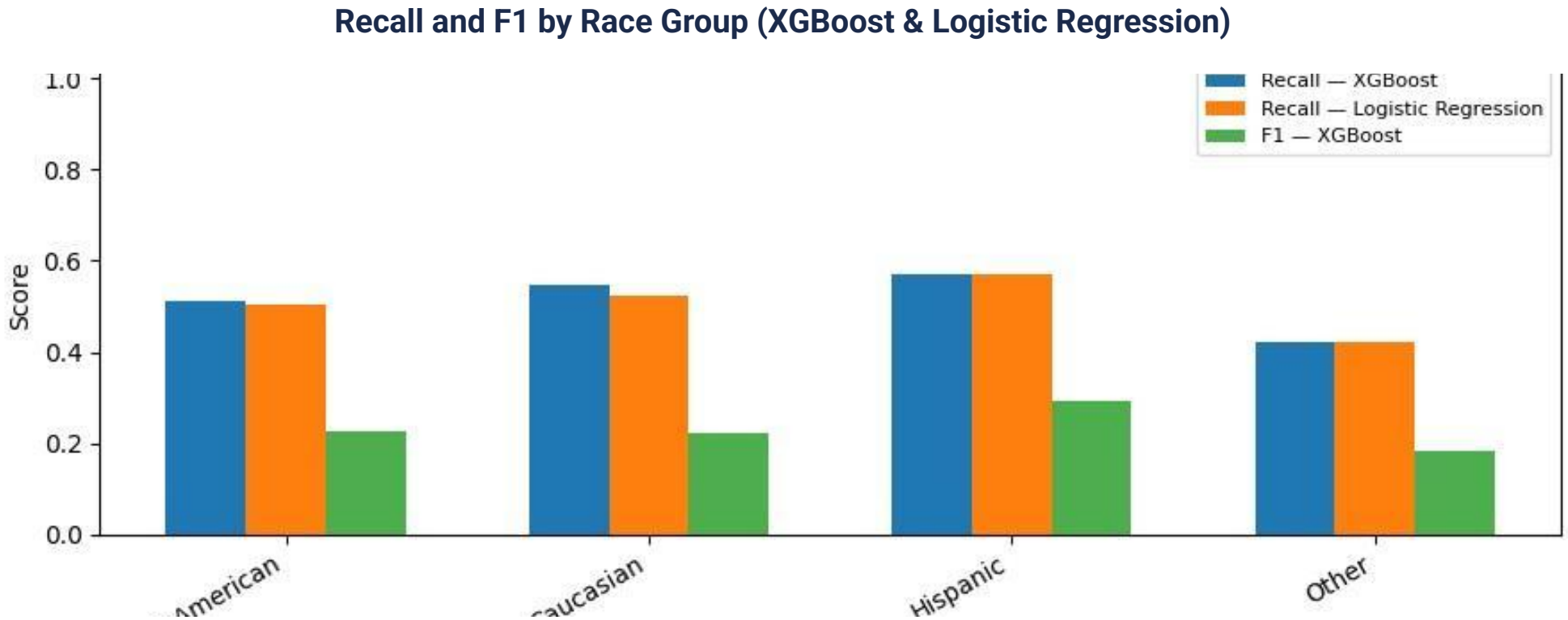
CONTEXT

The right threshold depends on hospital capacity and the cost of missing a readmission versus generating a false alarm.

Key Takeaway

While SVM required extreme threshold tuning (0.108), XGBoost was stable near 0.5. Threshold selection bridges the gap between statistical ranking (AUC) and clinical decision-making.

Fairness Analysis



Key Observations

The "Other" group shows lowest recall (0.421), indicating potential underservice. Gaps are consistent across both XGBoost and LR, suggesting the bias is data-driven rather than model-specific.

Discussion: What We Learned

1. The Overall Signal is Weak

With the best AUC at 0.65, 30-day hospital readmission is inherently difficult to predict from administrative data alone.

2. Complexity Yields Diminishing Returns

XGBoost outperformed the much simpler Logistic Regression by only 0.006 AUC, raising questions about deploying complex models.

3. Consistent Feature Agreement

Regardless of the algorithm, all models converged on prior utilization and discharge context as the most crucial predictors.

4. Thresholds Matter Deeply

In imbalanced classification contexts, explicitly tuning the decision threshold matters just as much as the initial model choice.

5. Deep Learning Underperformed

Deeper neural networks overfit; tabular data with handcrafted features is not the regime where representation learning shines.

Key Takeaway

Honest preprocessing, consistent evaluation, and ongoing fairness monitoring matter far more in practice than picking the fanciest modeling algorithm.

Study Limitations & Constraints

1. Historical Dataset constraints

Data spans 1999–2008. Clinical practices, treatment guidelines, and coding standards have significantly evolved since then.

2. Binary Target Simplification

The binary formulation collapses ">30 days" and "NO readmission" into a single class, losing important temporal granularity.

3. Survivorship Bias

Removing deceased and hospice-bound patients during preprocessing introduces survivorship bias into the modeling pipeline.

4. No External Validation

Models were strictly evaluated on the UCI test split. Performance generalizing to an entirely new hospital system remains untested.

5. Subjective Feature Engineering

Decisions such as how to group ICD-9 codes were subjective; different aggregation choices could yield different predictive signals.

6. Single Random Split Variance

Evaluation relies on a single 80/20 split (seed=42). Results might exhibit variance if tested across multiple independent random splits.

Key Takeaway

An honest assessment shows these models should serve as clinical decision aids, not autonomous decision-makers, due to data constraints.

Clinical Applicability: Deployment Recommendations

Which model would we deploy? It depends on the hospital's specific operational needs and regulatory constraints.

Auditable Requirements

If the hospital requires directly auditable coefficients for regulatory compliance or strict clinical oversight:

→ **Deploy: Logistic Regression**

Performance Focus

If the hospital prioritizes capturing complex interactions and accepts SHAP values for post-hoc monitoring:

→ **Deploy: XGBoost Ensemble**

Universal Prerequisites for Deployment

Regardless of the model chosen, successful clinical integration requires addressing key operational factors.

Critical Path

Either path MUST pair with: Tuned clinical thresholds, prospective validation, and ongoing fairness monitoring.

Conclusion

Best Performance

XGBoost achieved the best ranking performance (AUC 0.6505) on this 30-day readmission prediction task, consistent with published benchmarks.

Complexity vs. Value

The marginal gain over logistic regression (0.006 AUC) raises a legitimate question about whether the added complexity is justified for clinical deployment.

Rigorous Methodology

Honest preprocessing, consistent evaluation, and fairness monitoring matter more than picking the fanciest algorithm.